

Enhancing McAdams Coefficient-Based Speaker Anonymisation with Cross-Gender Timbre Transfer

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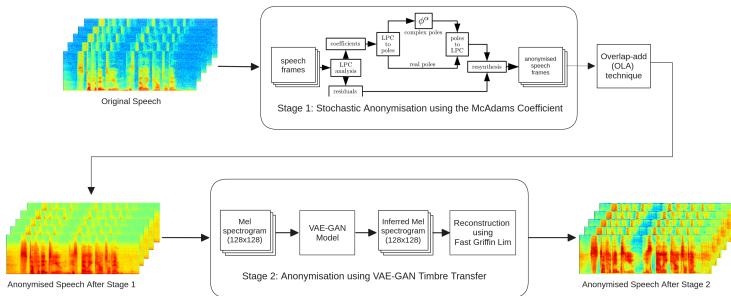
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Introduction: Speaker Anonymisation

- ① Modifying original speech to resemble the voice of a non-existent or unspecified speaker → privacy
- ① Preserving linguistic content and speech quality → utility

This work follows the VoicePrivacy2022 Challenge, aims to further explore the potential of dsp techniques and disguising gender id using a neural-style transfer model as a solution to the speaker anonymisation task.

System Overview



Speaker-Level Anonymisation & Randomised Gender Selection

System	source gender id	target gender id	level
Anonymisation	spk2gender	randomised	speaker
ASV (for eval)	spk2gender	randomised	utterance

Primary Privacy Evaluation: EER

Dataset	F/M	Weight	EER%				
			Orig	B1.a	B1.b	B2	Proposed
LibriSpeech-dev	F	0.25	8.67	17.76	19.03	11.36	11.93
	M	0.25	1.24	6.37	5.59	1.40	2.95
VCTK-dev (different)	F	0.20	2.86	12.46	8.25	6.68	11.29
	M	0.20	1.44	9.33	6.01	6.35	8.39
VCTK-dev (common)	F	0.05	2.62	13.95	9.01	5.81	7.27
	M	0.05	1.43	13.11	9.40	8.83	6.55
Weighted Average Dev Set			3.54	11.74	9.93	6.53	8.35
LibriSpeech-test	F	0.25	7.66	12.04	9.49	7.12	7.85
	M	0.25	1.11	8.91	7.80	1.11	2.00
VCTK-test (different)	F	0.20	4.89	16.00	10.91	16.92	13.68
	M	0.20	2.07	10.05	7.52	7.69	8.61
VCTK-test (common)	F	0.05	2.89	17.34	15.32	10.98	8.96
	M	0.05	1.13	9.89	8.19	4.80	5.93
Weighted Average Test Set			3.79	11.81	9.18	7.77	7.67

Table 3. Primary objective privacy evaluation results of metric EER% for ASV_{eval}^{anon} , compared with the original data and three baselines.

Primary Utility Evaluation: WER

Dataset	WER%				
	Orig	B1.a	B1.b	B2	Proposed
LibriSpeech-dev	3.82	4.34	4.19	4.32	5.88
VCTK-dev	10.79	11.54	10.98	11.76	15.05
Average Dev Set	7.31	7.94	7.59	8.04	10.465
LibriSpeech-test	4.15	4.75	4.43	4.58	5.92
VCTK-test	12.82	11.82	10.69	13.48	17.88
Average Test Set	8.49	8.29	7.56	9.03	11.90

Table 5. Primary objective utility evaluation results of metric WER% for ASR_{eval}^{anon} , compared with the original data and three baselines.

Secondary Utility Evaluation: p^{F_0} and G_{VD}

Dataset	F/M	Weight	p^{F_0}				G_{VD}			
			B1.a	B1.b	B2	Proposed	B1.a	B1.b	B2	Proposed
LibriSpeech-dev	F	0.25	0.77	0.84	0.64	0.29	-9.15	-4.92	-1.94	-1.36
	M	0.25	0.73	0.76	0.53	0.21	-8.94	-6.38	-1.65	-1.33
VCTK-dev (different)	F	0.20	0.84	0.87	0.70	0.36	-8.82	-5.94	-1.32	-1.38
	M	0.20	0.78	0.76	0.59	0.28	-12.61	-9.38	-2.18	-1.46
VCTK-dev (common)	F	0.05	0.79	0.84	0.64	0.29	-7.56	-4.17	-1.14	-0.99
	M	0.05	0.72	0.72	0.54	0.27	-10.37	-6.99	-1.32	-1.14
Weighted Average Dev Set			0.77	0.80	0.61	0.28	-9.71	-6.44	-1.72	-1.34
LibriSpeech-test	F	0.25	0.77	0.85	0.61	0.25	-10.04	-5.00	-1.71	-1.37
	M	0.25	0.69	0.72	0.54	0.22	-9.01	-6.64	-1.74	-1.30
VCTK-test (different)	F	0.20	0.84	0.87	0.68	0.28	-10.29	-6.09	-1.56	-3.64
	M	0.20	0.79	0.77	0.66	0.25	-11.69	-8.64	-1.56	-1.40
VCTK-test (common)	F	0.05	0.79	0.85	0.65	0.28	-9.31	-5.10	-1.59	-3.16
	M	0.05	0.70	0.71	0.61	0.26	-10.43	-6.50	-1.36	-1.10
Weighted Average Test Set			0.77	0.80	0.62	0.25	-10.15	-6.44	-1.63	-1.89

Table 7. Secondary objective utility evaluation results of metric pitch correlation p^{F_0} and gain of voice distinctiveness G_{VD} , compared with three baselines.

Evaluation Summary

- Enhanced privacy (higher EER) and greater voice distinctiveness are achieved compared to baseline systems.
- Proposed system lacks utility in terms of WER and pitch correlation.